

1. What is PHP?

PHP (Hypertext Preprocessor) is a **server-side scripting language** used to develop dynamic web applications.

It runs on server and sends output to browser

Works with HTML and databases

Used in:

- Web development
- Form handling
- Database operations

2. Basic PHP Syntax

```
<?php  
echo "Hello World";  
?>
```

Explanation:

- `<?php ?>` → PHP code block
- `echo` → prints output

Use:

Write PHP code inside HTML

3. Variables

```
$name = "John";  
$age = 20;
```

Explanation:

- \$ → used before variable
- No need to define data type

Use:

Store data

4. Data Types

```
$name = "John"; // String  
$age = 20;      // Integer  
$price = 10.5;  // Float  
$isTrue = true; // Boolean
```

Types:

- String
- Integer
- Float
- Boolean
- Array
- Object

Use:

Define data

5. Constants

```
define("PI", 3.14);
```

Use:

Fixed value

6. Operators

```
$sum = 5 + 3;
```

Types:

- Arithmetic
- Assignment
- Comparison
- Logical

Use:

Perform operations

7. Conditional Statements

```
if($age > 18) {  
    echo "Adult";  
} else {  
    echo "Child";  
}
```

Use:

Decision making

8. Switch Statement

```
switch($day) {  
  case 1: echo "Monday"; break;  
}
```

Use:

Multiple conditions

9. Loops

For Loop

```
for($i=0; $i<5; $i++) {  
  echo $i;  
}
```

While Loop

```
while($i < 5) {  
  $i++;  
}
```

Use:

Repeat code

10. Arrays

```
$arr = array(1,2,3);
```

Use:

Store multiple values

11. Associative Array

```
$person = array("name"=>"John", "age"=>20);
```

Use:

Key-value pairs

12. Functions

```
function greet() {  
    echo "Hello";  
}
```

Use:

Reusable code

13. String Functions

```
echo strlen("Hello");  
echo str_replace("H", "J", "Hello");
```

Use:

Text manipulation

14. Superglobals

```
$_GET  
$_POST  
$_SERVER
```

Use:

Access global data

15. Form Handling

```
$name = $_POST['name'];  
echo $name;
```

Use:

Get user input

16. File Handling

```
$file = fopen("file.txt", "r");
```

Use:

Read/write files

17. Sessions

```
session_start();  
$_SESSION['user'] = "John";
```

Use:

Store user data

18. Cookies

```
setcookie("user", "John", time()+3600);
```

Use:

Store data in browser

19. Database Connection (MySQL)

```
$conn = mysqli_connect("localhost","root","","db");
```

Use:

Connect database

20. Insert Data

```
mysqli_query($conn, "INSERT INTO users VALUES('John')");
```

Use:

Add data

21. Fetch Data

```
$result = mysqli_query($conn, "SELECT * FROM users");
```

Use:

Retrieve data

22. OOP in PHP

```
class Person {  
    public $name;  
}
```

Use:

Object-oriented programming

23. Constructor

```
function __construct() {  
    echo "Created";  
}
```

Use:

Initialize object

24. Inheritance

```
class B extends A {}
```

Use:

Code reuse

25. Exception Handling

```
try {  
    throw new Exception("Error");  
} catch(Exception $e) {  
    echo $e->getMessage();  
}
```

Use:

Handle errors

26. PHP Filters

```
filter_var($email, FILTER_VALIDATE_EMAIL);
```

Use:

Validate data

27. File Upload

```
move_uploaded_file($_FILES['file']['tmp_name'], "file.jpg");
```

Use:

Upload files

28. JSON in PHP

```
echo json_encode($arr);
```

Use:

Data exchange

29. Date & Time

```
echo date("Y-m-d");
```

Use:

Work with dates